

ADI-FDTD for Inhomogeneous Media

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1 Preliminaries

1.1 Maxwell's Equations

$$\nabla \times \mathbf{H} = \varepsilon \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E} \quad (1.1a)$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \quad (1.1b)$$

Expanding (1.1a) into components:

$$\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} = \varepsilon \frac{\partial E_x}{\partial t} + \sigma E_x \quad (1.2a)$$

$$\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} = \varepsilon \frac{\partial E_y}{\partial t} + \sigma E_y \quad (1.2b)$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = \varepsilon \frac{\partial E_z}{\partial t} + \sigma E_z \quad (1.2c)$$

Expanding (1.1b) into components:

$$\frac{\partial H_x}{\partial t} = \frac{1}{\mu} \left(\frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y} \right) \quad (1.3a)$$

$$\frac{\partial H_y}{\partial t} = \frac{1}{\mu} \left(\frac{\partial E_z}{\partial x} - \frac{\partial E_x}{\partial z} \right) \quad (1.3b)$$

$$\frac{\partial H_z}{\partial t} = \frac{1}{\mu} \left(\frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x} \right) \quad (1.3c)$$

1.2 Yee Grid in AMReX

The Yee grid stores different field components at different physical locations within each computational cell. In AMReX this staggering is represented by the `IndexType` used to define each `MultiFab`: a cell-centered direction has index type 0 and is located at $i + \frac{1}{2}$, while a nodal direction has index type 1 and is located at i . The implementation uses

$\mathbf{E}_\alpha$: cell-centered in the α direction and nodal in the other two directions,

$\mathbf{H}_\alpha$: nodal in the α direction and cell-centered in the other two directions.

Thus the component locations are

Component	x location	y location	z location
$E_{x,i,j,k}$	$i + \frac{1}{2}$	j	k
$E_{y,i,j,k}$	i	$j + \frac{1}{2}$	k
$E_{z,i,j,k}$	i	j	$k + \frac{1}{2}$
$H_{x,i,j,k}$	i	$j + \frac{1}{2}$	$k + \frac{1}{2}$
$H_{y,i,j,k}$	$i + \frac{1}{2}$	j	$k + \frac{1}{2}$
$H_{z,i,j,k}$	$i + \frac{1}{2}$	$j + \frac{1}{2}$	k

The half-cell offsets therefore do not appear explicitly as array indices such as $i + \frac{1}{2}$. They are encoded in the **MultiFab** staggering. The integer offsets in the finite-difference stencil choose the two source values that bracket the field being updated. For example, $H_{x,i,j,k}$ is located at $j + \frac{1}{2}$ in the y direction, while $E_{z,i,j,k}$ and $E_{z,i,j+1,k}$ are located at j and $j + 1$. Therefore

$$\left. \frac{\partial E_z}{\partial y} \right|_{H_{x,i,j,k}} \approx \frac{E_{z,i,j+1,k} - E_{z,i,j,k}}{\Delta y}. \quad (1.4)$$

Similarly, $E_{x,i,j,k}$ is located at j , while $H_{z,i,j-1,k}$ and $H_{z,i,j,k}$ are located at $j - \frac{1}{2}$ and $j + \frac{1}{2}$, so

$$\left. \frac{\partial H_z}{\partial y} \right|_{E_{x,i,j,k}} \approx \frac{H_{z,i,j,k} - H_{z,i,j-1,k}}{\Delta y}. \quad (1.5)$$

This is why the magnetic update below uses forward differences such as $j + 1$ and $k + 1$, while the electric update uses backward differences such as $j - 1$ and $k - 1$: each derivative is centered at the staggered location of the field component being updated.

2 FDTD Method

The explicit Yee FDTD update used here is written in three stages, matching the implementation: first update **H** by a half step, then update **E** by a full step, and finally update **H** by another half step. Define the update coefficients

$$C_a = \frac{2\varepsilon - \sigma\Delta t}{2\varepsilon + \sigma\Delta t}, \quad C_b = \frac{2\Delta t}{2\varepsilon + \sigma\Delta t}, \quad D_b = \frac{\Delta t}{\mu}, \quad (2.1)$$

where Δt is the time step, ε is the permittivity, μ is the permeability, and σ is the conductivity.

2.1 First Magnetic Half-Step

$$H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} = H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}}}{2} \left[\frac{E_{z,i,j+1,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n}{\Delta y} - \frac{E_{y,i,j+\frac{1}{2},k+1}^n - E_{y,i,j+\frac{1}{2},k}^n}{\Delta z} \right] \quad (2.2a)$$

$$H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} = H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - \frac{D_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{2} \left[\frac{E_{x,i+\frac{1}{2},j,k+1}^n - E_{x,i+\frac{1}{2},j,k}^n}{\Delta z} - \frac{E_{z,i+1,j,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n}{\Delta x} \right] \quad (2.2b)$$

$$H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} = H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - \frac{D_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}}}{2} \left[\frac{E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n}{\Delta x} - \frac{E_{x,i+\frac{1}{2},j+1,k}^n - E_{x,i+\frac{1}{2},j,k}^n}{\Delta y} \right]. \quad (2.2c)$$

In the AMReX Yee grid, with vacuum media, the magnetic field updates are implemented as

$$B_x(i, j, k) = B_x(i, j, k) - \frac{1}{2\Delta t} \left[\frac{E_z(i, j+1, k) - E_z(i, j, k)}{\Delta y} - \frac{E_y(i, j, k+1) - E_y(i, j, k)}{\Delta z} \right] \quad (2.3a)$$

$$B_y(i, j, k) = B_y(i, j, k) - \frac{1}{2\Delta t} \left[\frac{E_x(i, j, k+1) - E_x(i, j, k)}{\Delta z} - \frac{E_z(i+1, j, k) - E_z(i, j, k)}{\Delta x} \right] \quad (2.3b)$$

$$B_z(i, j, k) = B_z(i, j, k) - \frac{1}{2\Delta t} \left[\frac{E_y(i+1, j, k) - E_y(i, j, k)}{\Delta x} - \frac{E_x(i, j+1, k) - E_x(i, j, k)}{\Delta y} \right]. \quad (2.3c)$$

2.2 Electric Full-Step

$$E_{x,i+\frac{1}{2},j,k}^{n+1} = C_{a,i+\frac{1}{2},j,k}^{n+1} E_{x,i+\frac{1}{2},j,k}^n + C_{b,i+\frac{1}{2},j,k}^{n+1} \left[\frac{H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta y} - \frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} \right] \quad (2.4a)$$

$$E_{y,i,j+\frac{1}{2},k}^{n+1} = C_{a,i,j+\frac{1}{2},k}^{n+1} E_{y,i,j+\frac{1}{2},k}^n + C_{b,i,j+\frac{1}{2},k}^{n+1} \left[\frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} - \frac{H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta x} \right] \quad (2.4b)$$

$$E_{z,i,j,k+\frac{1}{2}}^{n+1} = C_{a,i,j,k+\frac{1}{2}}^{n+1} E_{z,i,j,k+\frac{1}{2}}^n + C_{b,i,j,k+\frac{1}{2}}^{n+1} \left[\frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y} \right]. \quad (2.4c)$$

In the AMReX Yee grid, with vacuum media, the electric field updates are implemented as

$$E_x(i, j, k) = E_x(i, j, k) + c^2 \Delta t \left[\frac{B_z(i, j, k) - B_z(i, j - 1, k)}{\Delta y} - \frac{B_y(i, j, k) - B_y(i, j, k - 1)}{\Delta z} \right] \quad (2.5a)$$

$$E_y(i, j, k) = E_y(i, j, k) + c^2 \Delta t \left[\frac{B_x(i, j, k) - B_x(i, j, k - 1)}{\Delta z} - \frac{B_z(i, j, k) - B_z(i - 1, j, k)}{\Delta x} \right] \quad (2.5b)$$

$$E_z(i, j, k) = E_z(i, j, k) + c^2 \Delta t \left[\frac{B_y(i, j, k) - B_y(i - 1, j, k)}{\Delta x} - \frac{B_x(i, j, k) - B_x(i, j - 1, k)}{\Delta y} \right]. \quad (2.5c)$$

2.3 Second Magnetic Half-Step

$$H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+1} = H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - \frac{D_b}{2} \left[\frac{E_{z,i,j+1,k+\frac{1}{2}}^{n+1} - E_{z,i,j,k+\frac{1}{2}}^{n+1}}{\Delta y} - \frac{E_{y,i,j+\frac{1}{2},k+1}^{n+1} - E_{y,i,j+\frac{1}{2},k}^{n+1}}{\Delta z} \right] \quad (2.6a)$$

$$H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+1} = H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - \frac{D_b}{2} \left[\frac{E_{x,i+\frac{1}{2},j,k+1}^{n+1} - E_{x,i+\frac{1}{2},j,k}^{n+1}}{\Delta z} - \frac{E_{z,i+1,j,k+\frac{1}{2}}^{n+1} - E_{z,i,j,k+\frac{1}{2}}^{n+1}}{\Delta x} \right] \quad (2.6b)$$

$$H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+1} = H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - \frac{D_b}{2} \left[\frac{E_{y,i+1,j+\frac{1}{2},k}^{n+1} - E_{y,i,j+\frac{1}{2},k}^{n+1}}{\Delta x} - \frac{E_{x,i+\frac{1}{2},j+1,k}^{n+1} - E_{x,i+\frac{1}{2},j,k}^{n+1}}{\Delta y} \right]. \quad (2.6c)$$

In the AMReX Yee grid, with vacuum media, the second magnetic field updates are implemented as

$$B_x(i, j, k) = B_x(i, j, k) - \frac{1}{2\Delta t} \left[\frac{E_z(i, j + 1, k) - E_z(i, j, k)}{\Delta y} - \frac{E_y(i, j, k + 1) - E_y(i, j, k)}{\Delta z} \right] \quad (2.7a)$$

$$B_y(i, j, k) = B_y(i, j, k) - \frac{1}{2\Delta t} \left[\frac{E_x(i, j, k + 1) - E_x(i, j, k)}{\Delta z} - \frac{E_z(i + 1, j, k) - E_z(i, j, k)}{\Delta x} \right] \quad (2.7b)$$

$$B_z(i, j, k) = B_z(i, j, k) - \frac{1}{2\Delta t} \left[\frac{E_y(i + 1, j, k) - E_y(i, j, k)}{\Delta x} - \frac{E_x(i, j + 1, k) - E_x(i, j, k)}{\Delta y} \right]. \quad (2.7c)$$

This explicit method is simple and local, but the time step is restricted by the CFL stability condition. For a uniform lossless grid, a typical stability limit is

$$\Delta t \leq \frac{1}{c \sqrt{\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} + \frac{1}{\Delta z^2}}}, \quad c = \frac{1}{\sqrt{\mu \varepsilon}}. \quad (2.8)$$

3 ADI Method

The ADI-FDTD update advances the fields by a full time step Δt in two half steps of size $\Delta t/2$. Unlike the explicit three-stage leapfrog update above, each ADI half step first updates $\mathbf{E}$ implicitly over $\Delta t/2$, then updates $\mathbf{H}$ explicitly over $\Delta t/2$ from the updated electric field. The first half step treats one set of implicit directions; the second half step treats the complementary directions to complete the cycle. Define the half-step update coefficients

$$C_a = \frac{4\varepsilon - \sigma\Delta t}{4\varepsilon + \sigma\Delta t}, \quad C_b = \frac{2\Delta t}{4\varepsilon + \sigma\Delta t}, \quad D_b = \frac{\Delta t}{2\mu}, \quad (3.1)$$

where Δt is the time step, ε is the permittivity, μ is the permeability, and σ is the conductivity. These differ from the full-step coefficients in (2.1) because each electric update uses $\Delta t/2$.

3.1 First ADI Half-Step

E_x and H_z By equation (1.2a),

$$\varepsilon_{i+\frac{1}{2},j,k} \frac{E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^n}{\Delta t/2} + \sigma_{i+\frac{1}{2},j,k} \frac{E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} + E_{x,i+\frac{1}{2},j,k}^n}{2} = [H_1], \quad (3.2)$$

where

$$[H_1] = \frac{H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y} - \frac{H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^n_{y,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z}. \quad (3.3)$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_x^{n+\frac{1}{2}} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_x^n + [H_1]. \quad (3.4)$$

Hence

$$E_x^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} = C_{a,i+\frac{1}{2},j,k} E^n_{x,i+\frac{1}{2},j,k} + C_{b,i+\frac{1}{2},j,k} [H_1], \quad (3.5)$$

By equation (1.3c),

$$H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^n_{z,i+\frac{1}{2},j+\frac{1}{2},k} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k} \left(\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta y} - \frac{E^n_{y,i+1,j+\frac{1}{2},k} - E^n_{y,i,j+\frac{1}{2},k}}{\Delta x} \right), \quad (3.6)$$

Substituting the magnetic update into (3.5) and eliminating $H_z^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} E_x^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} &= C_{a,i+\frac{1}{2},j,k} E^n_{x,i+\frac{1}{2},j,k} \\ &+ \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y} \left[H^n_{z,i+\frac{1}{2},j+\frac{1}{2},k} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k} \left(\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta y} - \frac{E^n_{y,i+1,j+\frac{1}{2},k} - E^n_{y,i,j+\frac{1}{2},k}}{\Delta x} \right) \right. \\ &\quad \left. - H^n_{z,i+\frac{1}{2},j-\frac{1}{2},k} - D_{b,i+\frac{1}{2},j-\frac{1}{2},k} \left(\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k}}{\Delta y} - \frac{E^n_{y,i+1,j-\frac{1}{2},k} - E^n_{y,i,j-\frac{1}{2},k}}{\Delta x} \right) \right] \\ &- \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z} \left(H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^n_{y,i+\frac{1}{2},j,k-\frac{1}{2}} \right). \end{aligned} \quad (3.7)$$

Collecting the unknown $E_x^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} &- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} \\ &+ \left[1 - \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2} + \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2} \right] E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} \\ &- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k} \\ &= C_{a,i+\frac{1}{2},j,k} E^n_{x,i+\frac{1}{2},j,k} + \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y} \left(H^n_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^n_{z,i+\frac{1}{2},j-\frac{1}{2},k} \right) \\ &- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y \Delta x} \left(E^n_{y,i+1,j+\frac{1}{2},k} - E^n_{y,i,j+\frac{1}{2},k} \right) \\ &+ \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y \Delta x} \left(E^n_{y,i+1,j-\frac{1}{2},k} - E^n_{y,i,j-\frac{1}{2},k} \right) \\ &- \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z} \left(H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^n_{y,i+\frac{1}{2},j,k-\frac{1}{2}} \right). \end{aligned} \quad (3.8)$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned}
-\alpha_{x1} E_{x,i+\frac{1}{2},j-1,k}^{n+\frac{1}{2}} + \beta_{x1} E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - \gamma_{x1} E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} &= p_{x1} E_{x,i+\frac{1}{2},j,k}^n \\
&+ \frac{1}{\Delta y} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^n \right) \\
&- \frac{1}{\Delta z} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^n \right) \\
&+ \frac{q_{x1}}{\Delta x \Delta y} \left(E_{y,i+1,j-\frac{1}{2},k}^n - E_{y,i,j-\frac{1}{2},k}^n \right) \\
&- \frac{r_{x1}}{\Delta x \Delta y} \left(E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n \right),
\end{aligned} \tag{3.9}$$

and the magnetic update can be written as

$$H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} = H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n + \frac{r_{x1}}{\Delta y} \left(E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \right) - \frac{r_{x1}}{\Delta x} \left(E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n \right). \tag{3.10}$$

where

$$\alpha_{x1} = \frac{D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}, \tag{3.11}$$

$$\beta_{x1} = \frac{1}{C_{b,i+\frac{1}{2},j,k}} + \alpha_{x1} + \gamma_{x1}, \tag{3.12}$$

$$\gamma_{x1} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2}, \tag{3.13}$$

$$p_{x1} = \frac{C_{a,i+\frac{1}{2},j,k}}{C_{b,i+\frac{1}{2},j,k}}, \tag{3.14}$$

$$q_{x1} = D_{b,i+\frac{1}{2},j-\frac{1}{2},k}, \tag{3.15}$$

$$r_{x1} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \tag{3.16}$$

E_y and H_x By equation (1.2b),

$$\varepsilon_{i,j+\frac{1}{2},k} \frac{E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^n}{\Delta t/2} + \sigma_{i,j+\frac{1}{2},k} \frac{E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} + E_{y,i,j+\frac{1}{2},k}^n}{2} = [H_2], \tag{3.17}$$

where

$$[H_2] = \frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} - \frac{H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^n}{\Delta x}. \tag{3.18}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2} \right) E_y^{n+\frac{1}{2}} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2} \right) E_y^n + [H_2]. \tag{3.19}$$

Hence

$$E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} = C_{a,i,j+\frac{1}{2},k} E_{y,i,j+\frac{1}{2},k}^n + C_{b,i,j+\frac{1}{2},k} [H_2], \tag{3.20}$$

By equation (1.3a),

$$\begin{aligned}
H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} &= H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \left(\frac{E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta z} \right. \\
&\quad \left. - \frac{E_{z,i,j+1,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n}{\Delta y} \right),
\end{aligned} \tag{3.21}$$

Substituting the magnetic update into (3.20) and eliminating $H_x^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned}
E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} &= C_{a,i,j+\frac{1}{2},k} E_{y,i,j+\frac{1}{2},k}^n \\
&+ \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z} \left[H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \left(\frac{E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta z} - \frac{E_{z,i,j+1,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n}{\Delta y} \right) \right. \\
&\quad \left. - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^n - D_{b,i,j+\frac{1}{2},k-\frac{1}{2}} \left(\frac{E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k-1}^{n+\frac{1}{2}}}{\Delta z} - \frac{E_{z,i,j+1,k-\frac{1}{2}}^n - E_{z,i,j,k-\frac{1}{2}}^n}{\Delta y} \right) \right] \\
&- \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^n \right).
\end{aligned} \tag{3.22}$$

Collecting the unknown $E_y^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned}
&- \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z^2} E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} \\
&+ \left[1 + \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z^2} + \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z^2} \right] E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \\
&- \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z^2} E_{y,i,j+\frac{1}{2},k-1}^{n+\frac{1}{2}} \\
&= C_{a,i,j+\frac{1}{2},k} E_{y,i,j+\frac{1}{2},k}^n + \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^n \right) \\
&- \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z \Delta y} \left(E_{z,i,j+1,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n \right) \\
&+ \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z \Delta y} \left(E_{z,i,j+1,k-\frac{1}{2}}^n - E_{z,i,j,k-\frac{1}{2}}^n \right) \\
&- \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^n \right).
\end{aligned} \tag{3.23}$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned}
-\alpha_{y1} E_{y,i,j+\frac{1}{2},k-1}^{n+\frac{1}{2}} + \beta_{y1} E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} - \gamma_{y1} E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} &= p_{y1} E_{y,i,j+\frac{1}{2},k}^n \\
&+ \frac{1}{\Delta z} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^n \right) \\
&- \frac{1}{\Delta x} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^n \right) \\
&+ \frac{q_{y1}}{\Delta y \Delta z} \left(E_{z,i,j+1,k-\frac{1}{2}}^n - E_{z,i,j,k-\frac{1}{2}}^n \right) \\
&- \frac{r_{y1}}{\Delta y \Delta z} \left(E_{z,i,j+1,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n \right),
\end{aligned} \tag{3.24}$$

and the magnetic update can be written as

$$H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} = H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n + \frac{r_{y1}}{\Delta z} \left(E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \right) - \frac{r_{y1}}{\Delta y} \left(E_{z,i,j+1,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n \right). \tag{3.25}$$

where

$$\alpha_{y1} = \frac{D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z^2}, \quad (3.26)$$

$$\beta_{y1} = \frac{1}{C_{b,i,j+\frac{1}{2},k}} + \alpha_{y1} + \gamma_{y1}, \quad (3.27)$$

$$\gamma_{y1} = \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z^2}, \quad (3.28)$$

$$p_{y1} = \frac{C_{a,i,j+\frac{1}{2},k}}{C_{b,i,j+\frac{1}{2},k}}, \quad (3.29)$$

$$q_{y1} = D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}, \quad (3.30)$$

$$r_{y1} = D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}. \quad (3.31)$$

E_z and H_y By equation (1.2c),

$$\varepsilon_{i,j,k+\frac{1}{2}} \frac{E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^n}{\Delta t/2} + \sigma_{i,j,k+\frac{1}{2}} \frac{E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} + E_{z,i,j,k+\frac{1}{2}}^n}{2} = [H_3], \quad (3.32)$$

where

$$[H_3] = \frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^n}{\Delta y}. \quad (3.33)$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2} \right) E_z^{n+\frac{1}{2}} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2} \right) E_z^n + [H_3]. \quad (3.34)$$

Hence

$$E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} = C_{a,i,j,k+\frac{1}{2}} E_{z,i,j,k+\frac{1}{2}}^n + C_{b,i,j,k+\frac{1}{2}} [H_3], \quad (3.35)$$

By equation (1.3b),

$$H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} = H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}} \left(\frac{E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{E_{x,i+\frac{1}{2},j,k+1}^n - E_{x,i+\frac{1}{2},j,k}^n}{\Delta z} \right), \quad (3.36)$$

Substituting the magnetic update into (3.35) and eliminating $H_y^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} &= C_{a,i,j,k+\frac{1}{2}} E_{z,i,j,k+\frac{1}{2}}^n \\ &+ \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x} \left[H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}} \left(\frac{E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{E_{x,i+\frac{1}{2},j,k+1}^n - E_{x,i+\frac{1}{2},j,k}^n}{\Delta z} \right) \right. \\ &\quad \left. - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^n - D_{b,i-\frac{1}{2},j,k+\frac{1}{2}} \left(\frac{E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i-1,j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{E_{x,i-\frac{1}{2},j,k+1}^n - E_{x,i-\frac{1}{2},j,k}^n}{\Delta z} \right) \right] \\ &- \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^n \right). \end{aligned} \quad (3.37)$$

Collecting the unknown $E_z^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned}
& - \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2} E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \\
& + \left[1 + \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2} + \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2} \right] E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \\
& - \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2} E_{z,i-1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \\
& = C_{a,i,j,k+\frac{1}{2}} E_{z,i,j,k+\frac{1}{2}}^n + \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^n \right) \\
& - \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x \Delta z} \left(E_{x,i+\frac{1}{2},j,k+1}^n - E_{x,i+\frac{1}{2},j,k}^n \right) \\
& + \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x \Delta z} \left(E_{x,i-\frac{1}{2},j,k+1}^n - E_{x,i-\frac{1}{2},j,k}^n \right) \\
& - \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^n \right).
\end{aligned} \tag{3.38}$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned}
& -\alpha_{z1} E_{z,i-1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} + \beta_{z1} E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - \gamma_{z1} E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} = p_{z1} E_{z,i,j,k+\frac{1}{2}}^n \\
& + \frac{1}{\Delta x} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^n \right) \\
& - \frac{1}{\Delta y} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^n \right) \\
& + \frac{q_{z1}}{\Delta z \Delta x} \left(E_{x,i-\frac{1}{2},j,k+1}^n - E_{x,i-\frac{1}{2},j,k}^n \right) \\
& - \frac{r_{z1}}{\Delta z \Delta x} \left(E_{x,i+\frac{1}{2},j,k+1}^n - E_{x,i+\frac{1}{2},j,k}^n \right),
\end{aligned} \tag{3.39}$$

and the magnetic update can be written as

$$H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} = H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n + \frac{r_{z1}}{\Delta x} \left(E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right) - \frac{r_{z1}}{\Delta z} \left(E_{x,i+\frac{1}{2},j,k+1}^n - E_{x,i+\frac{1}{2},j,k}^n \right). \tag{3.40}$$

where

$$\alpha_{z1} = \frac{D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}, \tag{3.41}$$

$$\beta_{z1} = \frac{1}{C_{b,i,j,k+\frac{1}{2}}} + \alpha_{z1} + \gamma_{z1}, \tag{3.42}$$

$$\gamma_{z1} = \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}, \tag{3.43}$$

$$p_{z1} = \frac{C_{a,i,j,k+\frac{1}{2}}}{C_{b,i,j,k+\frac{1}{2}}}, \tag{3.44}$$

$$q_{z1} = D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}, \tag{3.45}$$

$$r_{z1} = D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}. \tag{3.46}$$

In the AMReX Yee grid, with vacuum media, the field updates are implemented as

$$\begin{aligned}
-E_x(i, j-1, k) + \left(2 + \frac{4\Delta y^2}{c^2\Delta t^2}\right) E_x(i, j, k) - E_x(i, j+1, k) &= \frac{4\Delta y^2}{c^2\Delta t^2} E_x(i, j, k) \\
&+ \frac{2\Delta y^2}{\Delta t} \left[\frac{B_z(i, j, k) - B_z(i, j-1, k)}{\Delta y} - \frac{B_y(i, j, k) - B_y(i, j, k-1)}{\Delta z} \right] \\
&+ \Delta y \left[\frac{E_y(i+1, j-1, k) - E_y(i, j-1, k)}{\Delta x} - \frac{E_y(i+1, j, k) - E_y(i, j, k)}{\Delta x} \right], \tag{3.47a}
\end{aligned}$$

$$\begin{aligned}
-E_y(i, j, k-1) + \left(2 + \frac{4\Delta z^2}{c^2\Delta t^2}\right) E_y(i, j, k) - E_y(i, j, k+1) &= \frac{4\Delta z^2}{c^2\Delta t^2} E_y(i, j, k) \\
&+ \frac{2\Delta z^2}{\Delta t} \left[\frac{B_x(i, j, k) - B_x(i, j, k-1)}{\Delta z} - \frac{B_z(i, j, k) - B_z(i-1, j, k)}{\Delta x} \right] \\
&+ \Delta z \left[\frac{E_z(i, j+1, k-1) - E_z(i, j, k-1)}{\Delta y} - \frac{E_z(i, j+1, k) - E_z(i, j, k)}{\Delta y} \right], \tag{3.47b}
\end{aligned}$$

$$\begin{aligned}
-E_z(i-1, j, k) + \left(2 + \frac{4\Delta x^2}{c^2\Delta t^2}\right) E_z(i, j, k) - E_z(i+1, j, k) &= \frac{4\Delta x^2}{c^2\Delta t^2} E_z(i, j, k) \\
&+ \frac{2\Delta x^2}{\Delta t} \left[\frac{B_y(i, j, k) - B_y(i-1, j, k)}{\Delta x} - \frac{B_x(i, j, k) - B_x(i, j-1, k)}{\Delta y} \right] \\
&+ \Delta x \left[\frac{E_x(i-1, j, k+1) - E_x(i-1, j, k)}{\Delta z} - \frac{E_x(i, j, k+1) - E_x(i, j, k)}{\Delta z} \right], \tag{3.47c}
\end{aligned}$$

$$B_x(i, j, k) = B_x(i, j, k) + \frac{\Delta t}{2} \left[\frac{E_y(i, j, k+1) - E_y(i, j, k)}{\Delta z} - \frac{E_z(i, j+1, k) - E_z(i, j, k)}{\Delta y} \right], \tag{3.47d}$$

$$B_y(i, j, k) = B_y(i, j, k) + \frac{\Delta t}{2} \left[\frac{E_z(i+1, j, k) - E_z(i, j, k)}{\Delta x} - \frac{E_x(i, j, k+1) - E_x(i, j, k)}{\Delta z} \right], \tag{3.47e}$$

$$B_z(i, j, k) = B_z(i, j, k) + \frac{\Delta t}{2} \left[\frac{E_x(i, j+1, k) - E_x(i, j, k)}{\Delta y} - \frac{E_y(i+1, j, k) - E_y(i, j, k)}{\Delta x} \right]. \tag{3.47f}$$

3.2 Second ADI Half-Step

E_x and H_y By equation (1.2a),

$$\varepsilon_{i+\frac{1}{2}, j, k} \frac{E_{x, i+\frac{1}{2}, j, k}^{n+1} - E_{x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}}}{\Delta t/2} + \sigma_{i+\frac{1}{2}, j, k} \frac{E_{x, i+\frac{1}{2}, j, k}^{n+1} + E_{x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}}}{2} = [H_4], \tag{3.48}$$

where

$$[H_4] = \frac{H_{z, i+\frac{1}{2}, j+\frac{1}{2}, k}^{n+\frac{1}{2}} - H_{z, i+\frac{1}{2}, j-\frac{1}{2}, k}^{n+\frac{1}{2}}}{\Delta y} - \frac{H_{y, i+\frac{1}{2}, j, k+\frac{1}{2}}^{n+1} - H_{y, i+\frac{1}{2}, j, k-\frac{1}{2}}^{n+1}}{\Delta z}. \tag{3.49}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_x^{n+1} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_x^{n+\frac{1}{2}} + [H_4]. \tag{3.50}$$

Hence

$$E_{x, i+\frac{1}{2}, j, k}^{n+1} = C_{a, i+\frac{1}{2}, j, k} E_{x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}} + C_{b, i+\frac{1}{2}, j, k} [H_4], \tag{3.51}$$

By equation (1.3b),

$$\begin{aligned}
H_{y, i+\frac{1}{2}, j, k+\frac{1}{2}}^{n+1} &= H_{y, i+\frac{1}{2}, j, k+\frac{1}{2}}^{n+\frac{1}{2}} + D_{b, i+\frac{1}{2}, j, k+\frac{1}{2}} \left(\frac{E_{z, i+1, j, k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z, i, j, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} \right. \\
&\quad \left. - \frac{E_{x, i+\frac{1}{2}, j, k+1}^{n+1} - E_{x, i+\frac{1}{2}, j, k}^{n+1}}{\Delta z} \right), \tag{3.52}
\end{aligned}$$

Substituting the magnetic update into (3.51) and eliminating H_y^{n+1} terms gives

$$\begin{aligned}
E_{x,i+\frac{1}{2},j,k}^{n+1} &= C_{a,i+\frac{1}{2},j,k} E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \\
&+ \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\
&- \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z} \left[H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}} \left(\frac{E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{E_{x,i+\frac{1}{2},j,k+1}^{n+1} - E_{x,i+\frac{1}{2},j,k}^{n+1}}{\Delta z} \right) \right. \\
&\quad \left. - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}} - D_{b,i+\frac{1}{2},j,k-\frac{1}{2}} \left(\frac{E_{z,i+1,j,k-\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{E_{x,i+\frac{1}{2},j,k}^{n+1} - E_{x,i+\frac{1}{2},j,k-1}^{n+1}}{\Delta z} \right) \right]. \tag{3.53}
\end{aligned}$$

Collecting the unknown E_x^{n+1} terms gives

$$\begin{aligned}
&- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z^2} E_{x,i+\frac{1}{2},j,k+1}^{n+1} \\
&+ \left[1 + \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z^2} + \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z^2} \right] E_{x,i+\frac{1}{2},j,k}^{n+1} \\
&- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z^2} E_{x,i+\frac{1}{2},j,k-1}^{n+1} \\
&= C_{a,i+\frac{1}{2},j,k} E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} + \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\
&- \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}} \right) \\
&- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z \Delta x} \left(E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\
&+ \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z \Delta x} \left(E_{z,i+1,j,k-\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k-\frac{1}{2}}^{n+\frac{1}{2}} \right). \tag{3.54}
\end{aligned}$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned}
-\alpha_{x2} E_{x,i+\frac{1}{2},j,k-1}^{n+1} + \beta_{x2} E_{x,i+\frac{1}{2},j,k}^{n+1} - \gamma_{x2} E_{x,i+\frac{1}{2},j,k+1}^{n+1} &= p_{x2} E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \\
&+ \frac{1}{\Delta y} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\
&- \frac{1}{\Delta z} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}} \right) \\
&+ \frac{q_{x2}}{\Delta x \Delta z} \left(E_{z,i+1,j,k-\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k-\frac{1}{2}}^{n+\frac{1}{2}} \right) \\
&- \frac{r_{x2}}{\Delta x \Delta z} \left(E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right), \tag{3.55}
\end{aligned}$$

and the magnetic update can be written as

$$H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+1} = H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} + \frac{r_{x2}}{\Delta x} \left(E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right) - \frac{r_{x2}}{\Delta z} \left(E_{x,i+\frac{1}{2},j,k+1}^{n+1} - E_{x,i+\frac{1}{2},j,k}^{n+1} \right). \tag{3.56}$$

where

$$\alpha_{x2} = \frac{D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z^2}, \quad (3.57)$$

$$\beta_{x2} = \frac{1}{C_{b,i+\frac{1}{2},j,k}} + \alpha_{x2} + \gamma_{x2}, \quad (3.58)$$

$$\gamma_{x2} = \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z^2}, \quad (3.59)$$

$$p_{x2} = \frac{C_{a,i+\frac{1}{2},j,k}}{C_{b,i+\frac{1}{2},j,k}}, \quad (3.60)$$

$$q_{x2} = D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}, \quad (3.61)$$

$$r_{x2} = D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}. \quad (3.62)$$

E_y and H_z By equation (1.2b),

$$\varepsilon_{i,j+\frac{1}{2},k} \frac{E_{y,i,j+\frac{1}{2},k}^{n+1} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta t/2} + \sigma_{i,j+\frac{1}{2},k} \frac{E_{y,i,j+\frac{1}{2},k}^{n+1} + E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}}}{2} = [H_5], \quad (3.63)$$

where

$$[H_5] = \frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} - \frac{H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+1} - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^{n+1}}{\Delta x}. \quad (3.64)$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2} \right) E_y^{n+1} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2} \right) E_y^{n+\frac{1}{2}} + [H_5]. \quad (3.65)$$

Hence

$$E_{y,i,j+\frac{1}{2},k}^{n+1} = C_{a,i,j+\frac{1}{2},k} E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} + C_{b,i,j+\frac{1}{2},k} [H_5], \quad (3.66)$$

By equation (1.3c),

$$H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+1} = H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k} \left(\frac{E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_{y,i+1,j+\frac{1}{2},k}^{n+1} - E_{y,i,j+\frac{1}{2},k}^{n+1}}{\Delta x} \right), \quad (3.67)$$

Substituting the magnetic update into (3.66) and eliminating H_z^{n+1} terms gives

$$\begin{aligned} E_{y,i,j+\frac{1}{2},k}^{n+1} &= C_{a,i,j+\frac{1}{2},k} E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \\ &+ \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &- \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x} \left[H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k} \left(\frac{E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_{y,i+1,j+\frac{1}{2},k}^{n+1} - E_{y,i,j+\frac{1}{2},k}^{n+1}}{\Delta x} \right) \right. \\ &\quad \left. - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - D_{b,i-\frac{1}{2},j+\frac{1}{2},k} \left(\frac{E_{x,i-\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i-\frac{1}{2},j,k}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_{y,i,j+\frac{1}{2},k}^{n+1} - E_{y,i-1,j+\frac{1}{2},k}^{n+1}}{\Delta x} \right) \right]. \end{aligned} \quad (3.68)$$

Collecting the unknown E_y^{n+1} terms gives

$$\begin{aligned}
& -\frac{C_{b,i,j+\frac{1}{2},k} D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2} E_{y,i+1,j+\frac{1}{2},k}^{n+1} \\
& + \left[1 + \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2} + \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2} \right] E_{y,i,j+\frac{1}{2},k}^{n+1} \\
& - \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2} E_{y,i-1,j+\frac{1}{2},k}^{n+1} \\
& = C_{a,i,j+\frac{1}{2},k} E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} + \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^{n+\frac{1}{2}} \right) \\
& - \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\
& - \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x \Delta y} \left(E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \right) \\
& + \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x \Delta y} \left(E_{x,i-\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i-\frac{1}{2},j,k}^{n+\frac{1}{2}} \right).
\end{aligned} \tag{3.69}$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned}
-\alpha_{y2} E_{y,i-1,j+\frac{1}{2},k}^{n+1} + \beta_{y2} E_{y,i,j+\frac{1}{2},k}^{n+1} - \gamma_{y2} E_{y,i+1,j+\frac{1}{2},k}^{n+1} & = p_{y2} E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \\
& + \frac{1}{\Delta z} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^{n+\frac{1}{2}} \right) \\
& - \frac{1}{\Delta x} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\
& + \frac{q_{y2}}{\Delta y \Delta x} \left(E_{x,i-\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i-\frac{1}{2},j,k}^{n+\frac{1}{2}} \right) \\
& - \frac{r_{y2}}{\Delta y \Delta x} \left(E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \right),
\end{aligned} \tag{3.70}$$

and the magnetic update can be written as

$$H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+1} = H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} + \frac{r_{y2}}{\Delta y} \left(E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \right) - \frac{r_{y2}}{\Delta x} \left(E_{y,i+1,j+\frac{1}{2},k}^{n+1} - E_{y,i,j+\frac{1}{2},k}^{n+1} \right). \tag{3.71}$$

where

$$\alpha_{y2} = \frac{D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}, \tag{3.72}$$

$$\beta_{y2} = \frac{1}{C_{b,i,j+\frac{1}{2},k}} + \alpha_{y2} + \gamma_{y2}, \tag{3.73}$$

$$\gamma_{y2} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}, \tag{3.74}$$

$$p_{y2} = \frac{C_{a,i,j+\frac{1}{2},k}}{C_{b,i,j+\frac{1}{2},k}}, \tag{3.75}$$

$$q_{y2} = D_{b,i-\frac{1}{2},j+\frac{1}{2},k}, \tag{3.76}$$

$$r_{y2} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \tag{3.77}$$

E_z and H_x By equation (1.2c),

$$\varepsilon_{i,j,k+\frac{1}{2}} \frac{E_{z,i,j,k+\frac{1}{2}}^{n+1} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta t/2} + \sigma_{i,j,k+\frac{1}{2}} \frac{E_{z,i,j,k+\frac{1}{2}}^{n+1} + E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{2} = [H_6], \tag{3.78}$$

where

$$[H_6] = \frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+1} - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^{n+1}}{\Delta y}. \quad (3.79)$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_z^{n+1} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_z^{n+\frac{1}{2}} + [H_6]. \quad (3.80)$$

Hence

$$E_{z,i,j,k+\frac{1}{2}}^{n+1} = C_{a,i,j,k+\frac{1}{2}} E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} + C_{b,i,j,k+\frac{1}{2}} [H_6], \quad (3.81)$$

By equation (1.3a),

$$H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+1} = H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \left(\frac{E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta z} - \frac{E_{z,i,j+1,k+\frac{1}{2}}^{n+1} - E_{z,i,j,k+\frac{1}{2}}^{n+1}}{\Delta y} \right), \quad (3.82)$$

Substituting the magnetic update into (3.81) and eliminating H_x^{n+1} terms gives

$$\begin{aligned} E_{z,i,j,k+\frac{1}{2}}^{n+1} &= C_{a,i,j,k+\frac{1}{2}} E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \\ &+ \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &- \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y} \left[H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \left(\frac{E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta z} - \frac{E_{z,i,j+1,k+\frac{1}{2}}^{n+1} - E_{z,i,j,k+\frac{1}{2}}^{n+1}}{\Delta y} \right) \right. \\ &\quad \left. - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - D_{b,i,j-\frac{1}{2},k+\frac{1}{2}} \left(\frac{E_{y,i,j-\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j-\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta z} - \frac{E_{z,i,j,k+\frac{1}{2}}^{n+1} - E_{z,i,j-1,k+\frac{1}{2}}^{n+1}}{\Delta y} \right) \right]. \end{aligned} \quad (3.83)$$

Collecting the unknown E_z^{n+1} terms gives

$$\begin{aligned} &- \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2} E_{z,i,j+1,k+\frac{1}{2}}^{n+1} \\ &+ \left[1 + \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2} + \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2} \right] E_{z,i,j,k+\frac{1}{2}}^{n+1} \\ &- \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2} E_{z,i,j-1,k+\frac{1}{2}}^{n+1} \\ &= C_{a,i,j,k+\frac{1}{2}} E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} + \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &- \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &- \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y \Delta z} \left(E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\ &+ \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y \Delta z} \left(E_{y,i,j-\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j-\frac{1}{2},k}^{n+\frac{1}{2}} \right). \end{aligned} \quad (3.84)$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned}
-\alpha_{z2} E_{z,i,j-1,k+\frac{1}{2}}^{n+1} + \beta_{z2} E_{z,i,j,k+\frac{1}{2}}^{n+1} - \gamma_{z2} E_{z,i,j+1,k+\frac{1}{2}}^{n+1} &= p_{z2} E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \\
&+ \frac{1}{\Delta x} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\
&- \frac{1}{\Delta y} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\
&+ \frac{q_{z2}}{\Delta z \Delta y} \left(E_{y,i,j-\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j-\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\
&- \frac{r_{z2}}{\Delta z \Delta y} \left(E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \right),
\end{aligned} \tag{3.85}$$

and the magnetic update can be written as

$$H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+1} = H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} + \frac{r_{z2}}{\Delta z} \left(E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \right) - \frac{r_{z2}}{\Delta y} \left(E_{z,i,j+1,k+\frac{1}{2}}^{n+1} - E_{z,i,j,k+\frac{1}{2}}^{n+1} \right). \tag{3.86}$$

where

$$\alpha_{z2} = \frac{D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2}, \tag{3.87}$$

$$\beta_{z2} = \frac{1}{C_{b,i,j,k+\frac{1}{2}}} + \alpha_{z2} + \gamma_{z2}, \tag{3.88}$$

$$\gamma_{z2} = \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2}, \tag{3.89}$$

$$p_{z2} = \frac{C_{a,i,j,k+\frac{1}{2}}}{C_{b,i,j,k+\frac{1}{2}}}, \tag{3.90}$$

$$q_{z2} = D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}, \tag{3.91}$$

$$r_{z2} = D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}. \tag{3.92}$$

In the AMReX Yee grid, with vacuum media, the field updates are implemented as

$$\begin{aligned}
-E_x(i, j, k-1) + \left(2 + \frac{4\Delta z^2}{c^2 \Delta t^2} \right) E_x(i, j, k) - E_x(i, j, k+1) &= \frac{4\Delta z^2}{c^2 \Delta t^2} E_x(i, j, k) \\
&+ \frac{2\Delta z^2}{\Delta t} \left[\frac{B_z(i, j, k) - B_z(i, j-1, k)}{\Delta y} - \frac{B_y(i, j, k) - B_y(i, j, k-1)}{\Delta z} \right] \\
&+ \Delta z \left[\frac{E_z(i+1, j, k-1) - E_z(i, j, k-1)}{\Delta x} - \frac{E_z(i+1, j, k) - E_z(i, j, k)}{\Delta x} \right],
\end{aligned} \tag{3.93a}$$

$$\begin{aligned}
-E_y(i-1, j, k) + \left(2 + \frac{4\Delta x^2}{c^2 \Delta t^2} \right) E_y(i, j, k) - E_y(i+1, j, k) &= \frac{4\Delta x^2}{c^2 \Delta t^2} E_y(i, j, k) \\
&+ \frac{2\Delta x^2}{\Delta t} \left[\frac{B_x(i, j, k) - B_x(i, j, k-1)}{\Delta z} - \frac{B_z(i, j, k) - B_z(i-1, j, k)}{\Delta x} \right] \\
&+ \Delta x \left[\frac{E_x(i-1, j+1, k) - E_x(i-1, j, k)}{\Delta y} - \frac{E_x(i, j+1, k) - E_x(i, j, k)}{\Delta y} \right],
\end{aligned} \tag{3.93b}$$

$$\begin{aligned}
-E_z(i, j-1, k) + \left(2 + \frac{4\Delta y^2}{c^2 \Delta t^2} \right) E_z(i, j, k) - E_z(i, j+1, k) &= \frac{4\Delta y^2}{c^2 \Delta t^2} E_z(i, j, k) \\
&+ \frac{2\Delta y^2}{\Delta t} \left[\frac{B_y(i, j, k) - B_y(i-1, j, k)}{\Delta x} - \frac{B_x(i, j, k) - B_x(i, j-1, k)}{\Delta y} \right] \\
&+ \Delta y \left[\frac{E_y(i, j-1, k+1) - E_y(i, j-1, k)}{\Delta z} - \frac{E_y(i, j, k+1) - E_y(i, j, k)}{\Delta z} \right],
\end{aligned} \tag{3.93c}$$

$$B_x(i, j, k) = B_x(i, j, k) + \frac{\Delta t}{2} \left[\frac{E_y(i, j, k+1) - E_y(i, j, k)}{\Delta z} - \frac{E_z(i, j+1, k) - E_z(i, j, k)}{\Delta y} \right], \quad (3.93d)$$

$$B_y(i, j, k) = B_y(i, j, k) + \frac{\Delta t}{2} \left[\frac{E_z(i+1, j, k) - E_z(i, j, k)}{\Delta x} - \frac{E_x(i, j, k+1) - E_x(i, j, k)}{\Delta z} \right], \quad (3.93e)$$

$$B_z(i, j, k) = B_z(i, j, k) + \frac{\Delta t}{2} \left[\frac{E_x(i, j+1, k) - E_x(i, j, k)}{\Delta y} - \frac{E_y(i+1, j, k) - E_y(i, j, k)}{\Delta x} \right]. \quad (3.93f)$$